

**Humor Through Vision and Language: A Comparative Study of Cartoon Descriptions Using ViT and BLIP-2 and Joke Generation Using LLMs**

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## ABSTRACT

This paper presents a comparative study between Vision Transformer and BLIP-2 for generating image descriptions and extending these descriptions into humorous captions using LLMs. Image description generation is an important task in computer vision and multimodal artificial intelligence. In this I have tried to evaluate the performance of these models in their initial state and after fine tuning, using various metrics like BLEU score, ROUGE and word coverage. Then by defining which model performs better among the two I extend my study by using the generated description as input to multiple Large Language models(LLMs) to generate humorous captions or jokes for those images. Then I explore how different prompting techniques like one shot, few shot produce humorous captions. All in all I am trying to generate the creation of humorous captions, demonstrating the potential of multimodal AI in creative content generation. The results demonstrated that between ViT and BLIP2, BLIP 2 outperforms in generating descriptions for cartoon images and GPT 4 outperform all other LLMs. So a combination of BLIP-2 with GPT-4 achieves the best balance of creativity and relevance.

## 1. INTRODUCTION

Image Captioning bridges the gap between visual and textual models by generating descriptive sentences based on image content. The complexity increases when trying to extend captions to humor, requiring models to capture abstract concepts from a cartoon[14] image. This paper explores two challenges interconnected with each other.

First is Image Description Generation, in which I used two distinct approaches to image description. One is ViT, a transformer based architecture which divides images into patches and processes these patches to extract visual features. On the other hand, BLIP-2, uses pre-trained vision and language components for generating descriptions.

Second is generating humorous caption generation. Humor is an essential yet complex aspect of human creativity, which requires understanding of context, language, and cultural references. Generating humorous[28] captions for cartoons[14] presents unique challenges as we require understanding the abstract features of cartoon[14] images and generating humour.. Recent advancements in multimodal[19] models and LLMs have opened new avenues for exploring this domain, but understanding visual features from a cartoon and generating a caption that is humorous is still quite tricky.

In the previous studies that have been done on the New York style cartoons dataset all of them are based on training the model on the highest rated captions and image descriptions and then used to generate humorous caption, my aim here is to introduce a novel way in which we first understand the image, generate its description and then based on those descriptions generate jokes[29].

This study investigates the generation of humorous captions[27] for New Yorker-style cartoons by combining multimodal feature extractors, such as ViT and BLIP-2, with various LLMs for text generation. A comparative analysis is conducted on which vision model performs better in creating descriptions and on the effectiveness of different prompting strategies—zero-shot, one-shot, few-shot—and novel mechanisms like iterative refinement. I have also tried to extend my study by fine tuning my model but due to limited resources had a bit of constraint on that part.

## **1.1 Research Objectives**

1. Evaluate the performance of ViT and BLIP-2 in generating image descriptions.
2. Evaluating the impact of fine tuning on description quality
3. Investigate the capability of LLMs to generate humorous captions by giving the descriptions generated as input.
4. Assess the impact of different prompting techniques.

## **2. BACKGROUND**

The field of Artificial Intelligence(AI) has made remarkable strides in recent years, particularly in its ability to integrate visual and textual data. The progress has led to the development of models like Vision Transformers(ViT) and BLIP-2, which excel in multimodal tasks by combining vision and language processing. These advancements have opened new possibilities for applications such as image captioning, creative content generation, and contextual understanding of visual scenes.

Earlier we used to use convolutional neural networks(CNN[3]) and recurrent neural networks(RNNs) to extract features from images and generate text descriptions. But, these models had some limitations and faced challenges like limited ability to be able to capture global context and difficulties in training them as they used to face issues like vanishing gradients. So, I

decided to use more advanced models.

Instead of generating humor[28] directly from images, I thought of first making the model generate a literal description of the image. For that I have used two different image recognition[18] models.

## **2.1 Vision Transformer(ViT)**

In this we used attention[2] mechanisms to model relationships across all parts of an input sequence. Vision Transformer(ViT) introduced by Dosovitski et al. in 2020, represented a shift in image processing. It applies transformer[8] architecture originally developed for natural language processing to computer vision tasks. It is the state of the art[30] architecture vision recognition[3].

Architectural specifications:

- It divides images into fixed-size patches.
- It then linearly embeds these image patches
- On top of it applies standard transformer encoder mechanisms
- Then it utilized position embeddings to retain spatial information

Limitation of ViT:

- It has limited contextual understanding
- It may face challenges in capturing fine grained image details.
- It is dependent on large scale pre-training datasets.

## **2.2 BLIP-2: Bootstrapping Language-Image Pre-training**

Blip-2[4] was developed by Salesforce Research and it addressed many limitations of previous models through its innovative Querying Transformer(Q-Former) architecture. Its features mainly included advanced vision-language representation learning. It efficiently bridges between visual and linguistic domains. BLIP-2 integrates a pre-trained[10] vision encoder with language models which enables it to generate natural language descriptions based on the visual input it received.

This research is conducted in the context of the NYC Cartoon Dataset, which consists of New Yorker-style cartoons paired with captions. These cartoons are known for their unique blend of humor, cultural references, and visual storytelling. The dataset presents a challenge for AI systems because the captions must often be humorous, ironic, or absurd in a way that goes

beyond straightforward image description. This necessitates a deeper understanding of humor and its underlying mechanisms, which is why we are using BLIP-2[6] for generating descriptions and exploring how these descriptions can serve as input for large language models (LLMs) like GPT-2, GPT-4, LLaMA, and Mistral to generate humorous captions[27].

### **2.3 Large Language Models for Creative Content Generation**

This research explored four main Large Language Models:

1. GPT-2[13]
  - It was an early transformer based model
  - It has only foundational generative capabilities.
  - It returns limited contextual results as it is constrained by its size and training data.
  
2. LLaMa
  - It is an open- source model by Meta AI.
  - It has efficient computational performance
  - It has a improved and good context understanding
  
3. GPT-4[20]
  - It is an advanced multimodal language model, which can process both text and images.
  - It has state-of-the-art natural language understanding, because of which it is able to generate contextually appropriate tasks.
  
4. Mistral
  - It is also a open source model with a competitive performance with LLaMa
  - It is good for diverse language tasks, like, machine translation, summarization.

In this paper, I aim to check the effectiveness of ViT and BLIP-2 in generating descriptions for the NYC Cartoon Dataset and to research the use of LLMs for creating humorous[28] captions using prompt engineering.

### **3. RESEARCH AND METHODS**

#### **3.1 Dataset Preparation and Preprocessing**

In this research I used the NYC Cartoon Dataset from Hugging Face [1]. The dataset contains NYC style cartoons, with an elongated description of those images as well as captions for those images. For my research first I just used the images from the dataset to extract features and generate descriptions using two models BLIP-2[4] and ViT. I used the image\_description column of the dataset to serve as a benchmark for testing the comparative performance of BLIP-2 and ViT models in generating description of cartoon images. I then fine tuned my BLIP-2 model on a set of 500 cartoon images and descriptions and then retested and evaluated my results. Subsequently, I used the generated descriptions to generate jokes with the help of LLMs.

To use the images, I first converted all images to RGB format using the Python PIL library to ensure compatibility with ViT and BLIP-2[17] models. Then I expanded grayscale images to three channels and non-standard image shapes were resized to a uniform distribution (224\*224 pixels). Then in the code first I tokenized and cleaned captions by removing extraneous punctuation, special characters and redundant spaces. I also extracted metadata such as caption lengths and word frequencies to do analysis.

#### **3.2 Image Captioning Models**

As one of the main findings in [5], computer vision serves as a bottleneck for top-quality explanation generation; models that have access to human-labeled cartoon scene descriptions perform significantly better.

Vision Transformer(ViT) - This is a transformer based model. For this research, the ViT model that I used was nlpconnect/vit-gpt2-image-captioning. It was used to generate literal descriptions of an image. I tested its ability to encode visual information and decode captions. I did not perform any kind of fine tuning on ViT but used it as a strong comparison[11] with BLIP-2.

Bootstrapping Language-Image Pretraining(BLIP-2) - It is a vision model that was designed specifically for tasks like image captioning. It has a large pretrained encoder-decoder architecture that bridges the textual and visual modalities. For this research I used the model Salesforce/blip-image-captioning-large. I first used it as is without fine tuning and then as a next step I fine tuned it to have an enhanced understanding of the context in the cartoon images. The

fine tuned process implemented a custom Trainer class to ensure that there is compatibility with the dataset. In the training process I preprocessed cartoon images, tokenized the description and optimized performance using learning rate scheduler and Adam Optimizer. I used a custom trainer to train the model.

### 3.3 Evaluation Framework

To do the evaluation I created a custom class called ImageDescriptionEvaluator, so that I could compare the performance of ViT and BLIP-2 systematically in generating meaningful captions.

For evaluation I captured the description generated by ViT and Blip-2, and then compared those generated captions with the true descriptions given along with the cartoon images. To evaluate is used the below metrics:

ROUGE-1[22]- It measures the number of singular words that are overlapping between the generated text and reference text.

ROUGE-2[23]- It measures the overlap of two consecutive words also called bigrams between the generated text and reference text.

ROUGE-L[23]- It measures the longest subsequence that is common between the reference text and the generated text.

BLEU Score[25]: It basically evaluates the n-gram precision of the descriptions generated

Word-Coverage[26]: It evaluates the proportion of unique words from ground-truth statement that did occur in generated caption

### 3.4 Using Descriptions to Create Humorous Captions

The descriptions generated by the Image Captioning Models were then used as an input for various Large Language Models(LLMs) to generate humorous captions for the NYC Cartoon Dataset images. For this the study investigated how different prompting techniques performed. They are:

- **Zero-shot Prompting:** In this I just give a prompt like “Write a humorous caption for the following image description-> {image\_description}. I do not give any reference examples in this case.
- **One-shot Prompting:** In this a single example is provided of a humorous caption to guide the model’s response.

- **Few-shot Prompting[7]:** In this I gave multiple examples(max of 5 examples) to make the model better understand the context and tone.
- **Iterative Refinement Prompts:** With every response we ask it to improve on its response, i.e. it is an incremental method to generate captions to enhance humour and creativity.

### 3.5 Compare Different LLMs for Humor Generation

At last in this study, I used multiple LLMs namely, GPT-2, LLaMA, GPT-4 and Mistral for caption generation. I mainly used the descriptions generated by BLIP-2 as input to these models and then analyzed the humor of the captions that were generated. I evaluated them manually on criterias like -

The extent to which the caption had originality as well as was humorous.

Also, how well the generated caption aligned with the content in the cartoon.

## 4. EVALUATION AND RESULTS

Below I will explain how I reached my results step by step.

I have attached a NYC style cartoon Image below:





1: NYC style cartoon image

#### 4.1 Evaluations before fine tuning

In this case I used the Image Captioning models as is without training them on a specific dataset, so basically I use a pre fine tuned model. Firstly the above image is given as an input to the Image Captioning models.

##### ViT:

I used the Vision Transformer to get a description of the image. The ViT that is used was `nlpcconnect/vit-gpt2-image-captioning`

The **description Generated** by ViT is : "A painting of a painting of a man and a woman."

**Ground Truth Description:** "There are two men in suits standing at the door of an office both looking at a desk with a medieval sword stuck into the middle of the desk. One man is talking."

Then evaluated the above statement and compared it to the ground truth. The results that i received are:

##### ViT Metrics:

ROUGE-1: 0.22727272727272727

ROUGE-2: 0.0

ROUGE-L: 0.18181818181818182

BLEU Score[25]: 1.6879168804322106e-232

Word Coverage: 0.10714285714285714

**Observation:** Now we can observe that ViT is struggling to interpret nuanced scenes and anthropomorphism often seen in new york cartoons.

##### BLIP-2:

As a next test, I gave the same image as above as an input to BLIP-2 specifically using `Salesforce/blip-image-captioning-base`.

The **description generated** by this model is: "There is a man standing in front of a desk in an office"

**Ground Truth Description:** "There are two men in suits standing at the door of an office both looking at a desk with a medieval sword stuck into the middle of the desk. One man is talking."

Evaluation Metrics for Blip-2 model were:

**BLIP-2 Metrics:**

ROUGE-1: 0.47826086956521735

ROUGE-2: 0.09090909090909091

ROUGE-L: 0.21739130434782608

BLEU Score: 1.6828168187113343e-155

Word Coverage: 0.35714285714285715

**Observation:** In this case we can clearly observe that the captions are more context aware and detailed.

By comparing the generated description and evaluation metrics of above two models we can clearly state that BLIP-2 performed and demonstrated much better visual and contextual understanding when compared to ViT. BLIP-2 received higher ROUGE[24] scores and word coverage.

After experimenting with various cartoon images and receiving the results, I thought of improving these results by finetuning my model with images and description of images.

As Blip-2 clearly performed much better than ViT, I decided to just move forward with BLIP-2 and fine tune the BLIP-2 on the NYC cartoon dataset. So moving forward let's analyze the results by fine tuning the BLIP-2 model.

#### **4.2 Evaluations after Fine-Tuning BLIP-2 -**

So, I went forward and fine tuned BLIP-2 on 500 NYC style cartoon images and description of those images. Once I did that again I gave the above image as input and generated the literal description.

The **Generated Description** in this case is - "A man is standing in an office with a sword stuck into his desk, and another man looks at him angrily."

Now i evaluated the generated description with the ground truth, and got the below metrics:

**BLIP-2 Metrics:**

ROUGE-1: 0.5084745762711863

ROUGE-2: 0.21052631578947367

ROUGE-L: 0.3728813559322034

BLEU Score: 2.3236098643893214e-78

Word Coverage: 0.4642857142857143

**Observation:** We can observe that fine tuning enhanced BLIP-2's capabilities to capture the intricate details of cartoon images. The generated description is able to analyze all the characters and things in the image with minute details. After fine tuning the ROUGE score and word coverage also increased. This time it generated more specific and context-aware descriptions.

I tested my theory on various images and received good results. So, as I was able to generate literal descriptions from cartoon images, I decided to move my study a little further along by using the generated descriptions as input for generating humor using multiple LLMs and comparing the results generated content.

### 4.3 Humor Caption Generation:

I used the descriptions generated by BLIP-2, as input to the large language models to generate humorous captions under different prompt strategies. Below I have given sample example of some models and strategy that I used during the study:

#### Zero-shot prompting[16]->

Prompt - "write a humorous caption for the image whose description is given below. A man is standing in an office with a sword stuck into his desk, and another man looks at him angrily".

The responses generated by various large language model are as follow:

- **GPT-2-[13]**

"a painting of a painting of a man and a woman"

- **Llama** - (meta-llama/Meta-Llama-3.1-405B-Instruct-Turbo)-

"When you finally find a way to 'cut' through office bureaucracy, but your boss isn't impressed"

- **Mistral** - (Mistral-7B-Instruct-v0.2) -

"Looks like someone forgot 'Bring Your Sword to Work Day' was last Friday. Oops!

#MondayBlunders #DeskShenanigans" or "When you think you're the only one who can multitask with a sword... #OfficeDrama #SwordFu"

- **GPT-4** -

"When HR said to 'cut down on paperwork,' this probably wasn't what they meant."

**One shot Prompting**-> In this I used the captions as examples that are in the dataset from which I took the images and image\_description.

Prompt given to LLM is->

Description: "Two women are looking out a window. There is snow outside, and there is a snowman with human arms."

Joke: He has a summer job as a scarecrow.

Write a humorous caption for the image whose description is given below -

Description: a man is standing in an office with a sword stuck into his desk, and another man looks at him angrily.

- **GPT-2**

"When the printer jams for the third time this morning, drastic measures are taken."

- **Llama-**

"When you finally find a way to cut through the office bureaucracy."

- **Mistral-**

"I told him once 'bring me solutions, not problems' - but this is getting a little excessive!"

- **GPT-4 -**

"When you ask IT to *terminate* the virus, they take it a bit too literally."

Similar to above , I tested for a few shot prompting and the iterative prompting.

### **My Analysis-**

After a thorough examination of many diverse examples, my research yielded significant insights into AI model performance. GPT-4 and Llama emerged as the best performers, which consistently delivered great results. Llama demonstrated remarkable capabilities, giving a strong performance to GPT-4 and establishing itself as a good alternative.

Mistral[15] followed behind Llama, demonstrating mediocre performance. Among all of them, GPT-2 significantly underperformed when compared to other models even when advanced prompting techniques like few-shot and iterative prompting were applied.

I analyzed hundreds of examples and I would say that among all of the observed results, BLIP-2 and GPT-4 produced the most effective outcomes. Additionally, the prompting techniques that we used helped in reducing verbosity and generated more concise and targeted responses. So here we can clearly observe that model selection played an important role combined with prompting techniques that gave a good AI performance.

## **5. LIMITATIONS AND FUTURE WORK**

### **LIMITATIONS**

- The dataset size that I used was small because of limited resources.
- Biased humor assessment
- There may be same cultural and contextual bias

### **FUTURE RESEARCH DIRECTIONS**

- Humour generation between different cultures
- Creating better evaluation metrics for humor
- Creating multimodals that are adaptive to humor

## **6. CONCLUSION**

This comprehensive research illuminates the remarkable potential of multimodal AI in transforming visual information into creative and contextually rich textual content. By exploring image description generation and subsequent humor synthesis, this study is able to demonstrate the capabilities of AI systems.

This study helped in validating the effectiveness of BLIP-2 over ViT in the generation of cartoon image description. It also showed how there were significant improvements after fine tuning. Then we did research on how we could generate humor by different AI-prompting techniques.

In this study we can see how the rapidly evolving landscape of AI is able to highlight the interplay between vision[9] models and language models to generate meaningful and creative outputs.

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